Urysohn's Integral Equation

$$\boxed{x(s) + \int_{\Omega} k(s, t, x(t))dt = y(s)}$$

Generalization: kernel depends on solution

$k = k(s, t, x)$ (not just $k(s, t)$)

Key Features:

- Nonlinear integral equation
- Kernel depends on solution value
- Reduces to Hammerstein for $k = k(s, t) \cdot f(t, x(t))$
- Applications: physics, mechanics, biology